

Midterm 2021

Select the most appropriate answer(s) for each of the following questions. Put a circle around your choices:

- 1- The following data platform is an example of NoSQL databases:
 - a- PostgreSQL
 - b- Web Tier
 - c- Neo4J
 - d- MySQL
- 2- The key concept behind distributed system is that they are software applications to ensure there is a web tier for system administrators:
 - a- True
 - b- False
- 3- In relational database management systems, SQL is used to work and interact with the system. Which statement could be used to retrieve patient record given his age and location?
 - a- Insert..as Select..From
 - b- Select
 - c- Update
- 4- In distributed applications, two of the very key motivations to build them is:
 - a- Scalability to reduce down time
 - b- Availability to reduce down time
 - c- Scalability to cater for workload growth
 - d- Availability to cater for workload growth
- 5- The main difference of traditional database management systems and NoSQL databases is the way users login to
 - a- True
 - b- false
- 6- Distributed applications can be built in different topologies with key feature to enable an application to run over different physical or virtualized nodes. All the following are examples of such architecture except:
 - a- N-Tier Applications
 - b- Peer-to-Peer Applications
 - c- Node.js
 - d- Microservices

- 7- The main idea behind using public cloud is to pay as you utilize the services instead of buying and owning the infrastructure components needed for those services.
- a- True
 - b- False
- 8- The most important tier in Multi-Tier applications where the patient, hospitals, diagnostics, and other data are physically stored is:
- a- EHR
 - b- EPR
 - c- Database
 - d- AWS
- 9- When modeling the database for an EHR application, you wanted to ensure all the details of the data to be captured and stored is represented clearly. Which tool should be used:
- a- SQL
 - b- ERD
 - c- MongoDB
 - d- NoSQL
- 10- Amazon Web Services (AWS) is a collection of components including infrastructure, applications, databases, web servers among others offered in highly available, manageable, and integrated services.
- a- True
 - b- False
- 11- The following type of NoSQL database is best used to store and process data for an application that requires working with objects networked together:
- a- Redis Database
 - b- Graph Database
 - c- Hadoop HDFS
 - d- None of the above
- 12- The main difference in building distributed applications using: N-Tier and Microservices is:
- a- N-Tier require multiple nodes for application deployment
 - b- Microservices enable splitting the functionality of a single application
 - c- N-Tier can't be used for thick client models
 - d- Microservices enables the use of REST APIs and direct connectivity
- 13- The different functionality of applications when it comes to the nature of processing are:
- a- OLTP, OLAP and MDM
 - b- OLP, OLAP and Microservices
 - c- OLAP, OLTP and N-Tier
 - d- None of the above
- 14- ATM machines is one example of fat-client applications in the Client-Server architectural pattern.
- a- True
 - b- False

- 15- In relational database management systems, the ACID model is used for transaction control. What are the four features of this model?
- a- Atomicity, Consistency, Integration and Durability
 - b- Atomicity, Consistency, Isolation and Durability
 - c- Atomicity, Control, Isolation and Durability
 - d- Availability, Consistency, Isolation and Durability
- 16- Client-Server applications architecture include four layers. Following are three of them:
- a- Presentation
 - b- Data Handling
 - c- Transactions
 - d- Database
- 17- In distributed component architecture, the component that lays the central role in enabling other components to communicate together:
- a- Common Service
 - b- Client
 - c- Middleware
 - d- None of the above
- 18- BitTorrent and Bitcoin network are examples of the following distributed applications architecture:
- a- Multi-Tier Client-Server
 - b- Client-Server
 - c- Peer-to-Peer
 - d- Thin Client
- 19- Software as a service (SaaS) involves hosting the software remotely and providing access to it over the Internet:
- a- True
 - b- False
- 20- In relational database management systems, transactions are a group of commands that have to be executed and completed together. The following three commands could trigger a transaction:
- a- Select
 - b- Insert
 - c- Update
 - d- Delete
- 21- Electronic health Records (EHR) applications belongs to the which type of processing?
- a- OLTP
 - b- OLAP
 - c- MDM
 - d- None of the above
- 22- "How do you present each user of the software with the impression that they are working with their own copy of the system while, at the same time, making efficient use of system resources" called:
- a- Availability
 - b- Multi-tenancy
 - c- Service Configuration

- 23- Electronic health Records (EHR) applications main role is to provide replacement for other medical applications like HIS, PACT:
- a- True
 - b- False
- 24- Amazon Web Services (AWS) provides a way to manage relational databases on public cloud with cost effective way including running multiple DB instance. This service name is:
- a- AWS S3
 - b- AWS RDS
 - c- AWS Compute
 - d- None of the above
 - e- All of the above
- 25- The row and column way of representing and storing data is used for the following system:
- a- Key-value Stores
 - b- PostgreSQL
 - c- Web Servers
 - d- Graphs
- 26- Which of the following is an element of relational model for database management systems?
- a- Table
 - b- Row
 - c- Column
 - d- All of the above
- 27- The architecture of EHR doesn't require a database tier as it just connects to other medical applications:
- a- True
 - b- False
- 28- The row and column way of representing and storing data is used for the following systems:
- a- Key-value Stores
 - b- PostgreSQL
 - c- Web Servers
 - d- Graphs
- 29- A database system architecture contains only physical data storage on disks hence its role is to provide data management not memory management:
- a- True
 - b- False
- 30- When designing a database schema, ERD diagrams are used where the entities are representing:
- a- Rows
 - b- Columns
 - c- Tables
 - d- None of the above